

Overview

SkillsTrack is a client-driven web application built on HTML, CSS, and JavaScript in the browser, with Firebase supplying authentication and data storage as remote services. There is no custom backend server: the browser communicates directly with Firebase over the Firebase SDK and/or its REST API. The diagram below traces two concrete flows through the system — a learner signing in, and a learner creating a task — showing exactly where processing happens on the client versus the server, and how each request and response is handled.



User action



Client-side JS



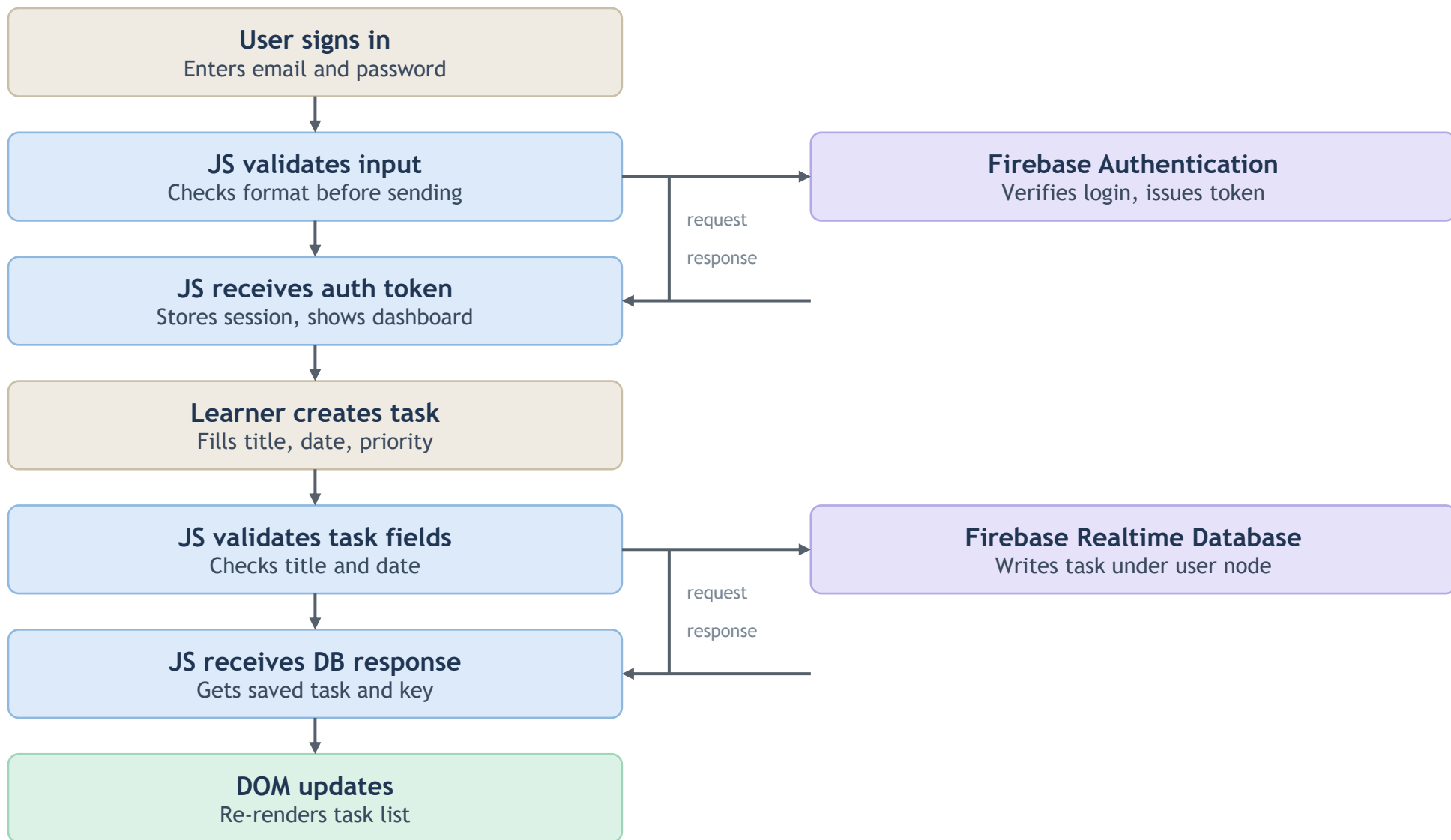
Firebase service



DOM update

Browser / client side

Firebase (remote services)



Flow 1 — Learner sign-in

The learner enters an email and password into the sign-in form in the browser.

Client-side JavaScript validates the input (checks that fields are non-empty and correctly formatted) before anything is sent over the network.

JavaScript sends an authentication request through the Firebase SDK (or REST API) to Firebase Authentication.

Firebase Authentication is a remote service: it verifies the credentials against its user store and issues a session token.

The token is returned in the response to the browser.

JavaScript receives the response, stores the session (e.g. in memory or local state), and updates the DOM to show the learner's dashboard.

Flow 2 — Creating a task

The learner fills out a task form: title, due date, and priority.

JavaScript validates the task fields client-side (e.g. title not empty, date is valid) before sending anything.

A write request is sent through the Firebase SDK/REST API to the Firebase Realtime Database, targeting the current user's node.

The Firebase Realtime Database, a remote data store, persists the new task and generates a unique key for it.

Firebase returns a response confirming the write, including the saved task data and its new key.

JavaScript receives that data and updates the DOM — the new task is rendered into the task list the learner sees, with no page reload.

Layer responsibilities summary

Layer	Runs where	Responsibility
HTML / CSS / JS	Client (browser)	Renders UI, captures user input, validates data before sending, updates the DOM when data returns.
Firebase SDK / REST API	Client → boundary	Serializes requests, carries them to Firebase, and returns responses/data to JavaScript.
Firebase Authentication	Remote (Firebase)	Verifies credentials, issues and validates session tokens.
Firebase Realtime Database	Remote (Firebase)	Stores and retrieves task data, keyed per user, and syncs changes back to the client.